Solution Starting time 22.30, Travelling time 32 hours. Here we use modulo 24.

The reaching time is

$$22.30 + 32 \pmod{24} \equiv 54.30 \pmod{24}$$
$$\equiv 6.30 \pmod{24} \quad (\because 32 = (1\times 24) + 8$$
Thursday Friday)

Thus, he will reach Delhi on Friday at 6.30 hours.

Example 2.18 Kala and Vani are friends. Kala says, "Today is my birthday" and she asks Vani, "When will you celebrate your birthday?" Vani replies, "Today is Monday and I celebrated my birthday 75 days ago". Find the day when Vani celebrated her birthday.

Solution Let us associate the numbers 0, 1, 2, 3, 4, 5, 6 to represent the weekdays from Sunday to Saturday respectively.

Vani says today is Monday. So the number for Monday is 1. Since Vani's birthday was 75 days ago, we have to subtract 75 from 1 and take the modulo 7, since a week contain 7 days.

$$-74 \pmod 7 \equiv -4 \pmod 7 \equiv 7-4 \pmod 7 \equiv 3 \pmod 7$$
$$(\because -74 - 3 = -77 \text{ is divisible by } 7)$$

Thus, $1 - 75 \equiv 3 \pmod 7$

The day for the number 3 is Wednesday.

Therefore, Vani's birthday must be on Wednesday.

Exercise 2.3

1. Find the least positive value of x such that
   (i) $71 \equiv x \pmod 8$ (ii) $78 + x \equiv 3 \pmod 5$ (iii) $89 \equiv (x+3) \pmod 4$
   (iv) $96 \equiv \frac{x}{7} \pmod 5$ (v) $5x \equiv 4 \pmod 6$
2. If x is congruent to 13 modulo 17 then $7x - 3$ is congruent to which number modulo 17?
3. Solve $5x \equiv 4 \pmod 6$
4. Solve $3x - 2 \equiv 0 \pmod{11}$
5. What is the time 100 hours after 7 a.m.?
6. What is the time 15 hours before 11 p.m.?
7. Today is Tuesday. My uncle will come after 45 days. In which day my uncle will be coming?
8. Prove that $2^n + 6 \times 9^n$ is always divisible by 7 for any positive integer n.
9. Find the remainder when 2^{81} is divided by 17.

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